



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Autodesk Certified Professional (ACP) - AutoCAD Design and Drafting

TGS Ref No: TGS-2023036004

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 9.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	14 June 2023	First version	Tertiary Infotech Academy
2.0	4 September 2024	Second version	Tertiary Infotech Academy
3.0		New format	Tertiary Infotech Academy
4.0	17 October 2025	Updated company name	Tertiary Infotech Academy
9.0	22 August 2026	Expanded four-day schedule aligned to the 363-slide v9 deck, all 53 December 2025 Autodesk objectives and sixteen self-contained AutoCAD labs.	Tertiary Infotech Academy

TABLE OF CONTENTS

Course Information	4
Learning Outcomes	4
Assessment	4
Course Schedule	4
Day 1 - Standards, drawing management and advanced geometry.....	4
Day 2 - Precision editing, reusable content and annotation.....	6
Day 3 - Layouts, publishing and collaboration.....	7
Day 4 - Integrated capstone, evidence review and assessment.....	8
Alignment Map	8

Course Information

Course title: Autodesk Certified Professional (ACP) - AutoCAD Design and Drafting

Course code: TGS-2023036004

TSC: Design Sketching · BEV-DES-5026-1.1-1

Duration: 4 days · 32 contact hours, including 3 hours of summative assessment

Mode: Instructor-led concept briefing, AutoCAD demonstration, individual lab practice, evidence review and assessment

Trainer: Dr Alfred Ang

Learning Outcomes

- LO1: Define the application of emerging design visualization standards in AutoCAD art direction and sketching.
- LO2: Develop the organisation's drawing and sketching capabilities using methods for revising 2D and 3D technical drawings.
- LO3: Develop organisation-specific drawing standards that use storytelling and annotation to engage stakeholders.
- LO4: Lead the integration of emerging drawing technologies and suitable methods in alignment with organisational growth.
- LO5: Inspire stakeholders to explore traditional and contemporary approaches to drawing and sketching.

Assessment

- Written Assessment (WA-SAQ) — 4 open-ended questions covering K1–K4, 60 minutes, open book.
- Practical Performance (PP) — 5 hands-on tasks covering A1–A5, 120 minutes, open book.
- Learners must demonstrate every required K and A criterion to achieve Competent (C).

Course Schedule

Day 1 - Standards, drawing management and advanced geometry

Time	Duration	Block	Content / activity	Slides
9:00-9:30	30 min	Orientation	Welcome, attendance, introductions, learning outcomes, TSC, Autodesk exam map and assessment briefing.	Slides 2-16

Time	Duration	Block	Content / activity	Slides
9:30-10:45	75 min	Domain 1A	Exam groups 1.1-1.2: layers, viewport overrides, filters, xref layer ownership and named UCS control.	Slides 20-45
10:45-11:00	15 min	Break	Tea break.	—
11:00-12:00	60 min	Lab 1	Drawing Standards and Project Setup; individual AutoCAD practice and evidence review.	Slide 100 · Lab 1 · LG Lab 1
12:00-13:00	60 min	Lunch	Lunch break.	—
13:00-14:15	75 min	Domain 1B	Exam groups 1.3-1.5: properties, measurement, COUNT, interface control, audit, recovery and purge decisions.	Slides 46-80
14:15-15:00	45 min	Lab 2	Layers, Object Management and Revision Control.	Slide 101 · Lab 2 · LG Lab 2
15:00-15:15	15 min	Break	Tea break.	—
15:15-16:15	60 min	Foundations	Technical drawing foundations for application menus, workspaces, palettes, units, navigation, properties and layers.	Slides 81-91
16:15-17:15	60 min	Lab 11	Drawing Environment, UCS and Recovery using the selected Technical Drawing and Autodesk datasets.	Slide 102 · Lab 11 · LG Lab 11
17:15-18:00	45 min	Review	Domain 1 evidence critique, objective traceability review and PM attendance.	Slides 93-102

Day total: 480 contact minutes (8 hours), excluding lunch and including scheduled tea breaks.

Day 2 - Precision editing, reusable content and annotation

Time	Duration	Block	Content / activity	Slides
9:00-9:15	15 min	Review	AM attendance and Day 1 retrieval practice.	Selected Day 1 slides
9:15-10:30	75 min	Domain 2A	Exam groups 2.1-2.2: annotation styles, drawing scales, assigned scales and scale-specific positions.	Slides 106-131
10:30-10:45	15 min	Break	Tea break.	—
10:45-11:30	45 min	Lab 3	Annotation Standards and Stakeholder Story.	Slide 155 · Lab 3 · LG Lab 3
11:30-12:15	45 min	Domain 2B	Exam group 2.3 and technical foundations: annotations, tables, hatches, multileaders and revision communication.	Slides 132-146
12:15-13:15	60 min	Lunch	Lunch break.	—
13:15-14:00	45 min	Lab 4	Dimensions, Multileaders and Data Tables.	Slide 156 · Lab 4 · LG Lab 4
14:00-14:45	45 min	Lab 12	Annotation Styles, Scales and Schedules using Technical Drawing component assets.	Slide 157 · Lab 12 · LG Lab 12
14:45-15:00	15 min	Break	Tea break.	—
15:00-16:15	75 min	Domain 3A	Exam groups 3.1-3.2: reusable content, clipboard control, arcs, polylines, offset, boundaries and arrays.	Slides 161-198
16:15-17:00	45 min	Lab 5	Advanced Geometry and Precision Editing.	Slide 251 · Lab 5 · LG Lab 5
17:00-17:45	45 min	Lab 13	Coordinate Geometry and Construction Objects.	Slide 253 · Lab 13 · LG Lab 13
17:45-18:00	15 min	Review	Day 2 evidence review and PM attendance.	Selected Day 2 slides

Day total: 480 contact minutes (8 hours), excluding lunch and including scheduled tea breaks.

Day 3 - Layouts, publishing and collaboration

Time	Duration	Block	Content / activity	Slides
9:00-9:15	15 min	Review	AM attendance and Day 2 retrieval practice.	Selected Day 2 slides
9:15-10:30	75 min	Domain 3B	Exam groups 3.3-3.4 and foundations: grips, dynamic input, snaps, coordinates and controlled modification.	Slides 199-237
10:30-10:45	15 min	Break	Tea break.	—
10:45-11:30	45 min	Lab 6	Blocks, Attributes and Data Extraction.	Slide 252 · Lab 6 · LG Lab 6
11:30-12:15	45 min	Lab 14	Associative Arrays and Grip-Based Revision.	Slide 254 · Lab 14 · LG Lab 14
12:15-13:15	60 min	Lunch	Lunch break.	—
13:15-14:30	75 min	Domain 4A	Exam group 4.1: named page setups, layout viewports, clipping, scales, locks and annotation control.	Slides 258-270
14:30-15:15	45 min	Lab 7	Layouts, Viewports and Page Setups using the sink dataset.	Slide 299 · Lab 7 · LG Lab 7
15:15-15:30	15 min	Break	Tea break.	—
15:30-16:30	60 min	Domain 4B	Exam groups 4.2-4.3: publish lists, eTransmit, sheet creation, subsets and sheet-set properties.	Slides 271-288
16:30-17:15	45 min	Lab 8	Publish, Plot and eTransmit.	Slide 300 · Lab 8 · LG Lab 8
17:15-18:00	45 min	Lab 15	Named Page Setups, Sheet Sets and Batch Output.	Slide 301 · Lab 15 · LG Lab 15

Day total: 480 contact minutes (8 hours), excluding lunch and including scheduled tea breaks.

Day 4 - Integrated capstone, evidence review and assessment

Time	Duration	Block	Content / activity	Slides
9:00-9:15	15 min	Review	AM attendance and Day 3 retrieval practice.	Selected Day 3 slides
9:15-10:15	60 min	Domain 5A	Exam groups 5.1-5.2: shared review, Traces, Compare, xref types, paths and underlay frames.	Slides 305-330
10:15-10:30	15 min	Break	Tea break.	—
10:30-11:15	45 min	Lab 9	Xrefs, Underlays and Drawing Comparison.	Slide 353 · Lab 9 · LG Lab 9
11:15-12:15	60 min	Domain 5B	Exam group 5.3 and collaboration foundations: PDF imports, data links, fields and controlled handover.	Slides 331-351
12:15-13:15	60 min	Lunch	Lunch break.	—
13:15-14:00	45 min	Lab 16	Shared Review, References and Linked Data.	Slide 355 · Lab 16 · LG Lab 16
14:00-14:30	30 min	Lab 10	Integrated Sink Documentation Capstone readiness and evidence handover.	Slide 354 · Lab 10 · LG Lab 10
14:30-14:45	15 min	Break	Tea break.	—
14:45-15:00	15 min	Assessment	Course synthesis, final assessment briefing and assessment digital attendance.	Slides 357-16
15:00-16:00	60 min	WA-SAQ	Four open-ended questions covering K1-K4.	WA question paper
16:00-18:00	120 min	PP	Five practical tasks covering A1-A5 using the sink-layout dataset; feedback and sign-off.	PP question paper · AutoCAD dataset

Day total: 480 contact minutes (8 hours), excluding lunch and including scheduled tea breaks.

Alignment Map

- Domain 1 and Labs 1, 2 and 11 -> LO1-LO2 · K1-K2 · A1-A2.
- Domain 2 and Labs 3, 4 and 12 -> LO3 · K3 · A3.
- Domain 3 and Labs 5, 6, 13 and 14 -> LO1-LO4 · K1, K2, K4 · A1, A2, A4.
- Domain 4 and Labs 7, 8 and 15 -> LO3-LO4 · K3, K4 · A1-A4.
- Domain 5 and Labs 9, 10 and 16 -> LO1-LO5 · K1-K4 · A1-A5.
- WA questions 1-4 -> K1-K4; PP tasks 1-5 -> A1-A5.